

6-Month AI/ML Roadmap (Dated, with Links + Tasks + Outputs)

(Start: Sept 1, 2025 – End: Feb 28, 2026)

Phase 1 (Sept 1 – Oct 26, 2025): Math Foundations (8 weeks)

Goal: Build ML math intuition + coding practice

◆ Week 1–2 (Sept 1–14): Linear Algebra

- **Courses:**
 - DataCamp – Linear Algebra for Data Science
 - [Coursera – Mathematics for Machine Learning: Linear Algebra](#)
- **Tasks:**
 - NumPy matrix ops (dot, transpose, inverse)
 - PCA decomposition with `sklearn`
- **Output:** Notebook: “*PCA from scratch + sklearn*”

◆ Week 3–4 (Sept 15–28): Calculus & Optimization

- **Course:** [Coursera – Calculus for Machine Learning & Data Science](#)
- **Tasks:**
 - Implement gradient descent from scratch
 - Plot convergence curves
- **Output:** Notebook: “*Gradient Descent Visualization*”

◆ Week 5–6 (Sept 29 – Oct 12): Probability & Distributions

- **Course:** [Coursera – Probability & Statistics for Machine Learning & Data Science](#)
- **Tasks:**
 - Apply Bayes’ theorem on toy dataset
 - Simulate Normal, Bernoulli, Poisson distributions
- **Output:** Notebook: “*Probability Playground*”

◆ Week 7 (Oct 13–19): Statistics & Hypothesis Testing

- **Course:** DataCamp – Foundations of Statistics for Machine Learning
- **Tasks:**
 - Run an A/B test (conversion dataset)
 - Hypothesis test using `pandas`
- **Output:** Notebook: “*A/B Test Analysis*”

◆ Week 8 (Oct 20–26): Math → ML Bridge

- **Task:** Write doc: “*Math inside Linear Regression, K-Means, Naive Bayes*”
- **Output:** Short PDF (3–5 pages)

✓ Deliverables (Phase 1):

- 3 coding notebooks (PCA, Gradient Descent, Probability/A-B Test)
 - 1 written doc
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Phase 2 (Oct 27 – Dec 21, 2025): Core ML/DL/NLU (8 weeks)

Goal: Build ML → DL → NLU projects

◆ Week 9 (Oct 27 – Nov 2): Supervised ML

- **Course:** DataCamp – Supervised Learning with scikit-learn
- **Task:** Kaggle Titanic competition ([link](#))
- **Output:** Notebook + Kaggle submission

◆ Week 10 (Nov 3–9): Unsupervised ML

- **Course:** DataCamp – Unsupervised Learning in Python
- **Task:** K-Means + PCA visualization on dataset
- **Output:** Notebook: “*Unsupervised Learning Demo*”

◆ Week 11–12 (Nov 10–23): Neural Networks + Backprop

- **Course:** [Coursera – Neural Networks & Deep Learning](#)
- **Tasks:**
 - Build 2-layer NN in NumPy
 - Implement backprop step-by-step
- **Output:** Notebook: “*2-Layer Neural Net from Scratch*”

◆ Week 13 (Nov 24–30): CNNs

- **Course:** [Coursera – Convolutional Neural Networks](#)
- **Task:** Build CNN for MNIST (Keras/TensorFlow)
- **Output:** Notebook: “*MNIST CNN Classifier*”

◆ Week 14 (Dec 1–7): RNNs & LSTMs

- **Course:** [Coursera – Sequence Models](#)
- **Task:** RNN text generator (e.g., Shakespeare text)
- **Output:** Notebook: “*RNN Text Generator*”

◆ Week 15 (Dec 8–14): Transformers

- **Course:** HuggingFace Transformers Course
- **Task:** Sentiment analysis with pre-trained BERT
- **Output:** Notebook: “*Sentiment Analysis with BERT*”

◆ Week 16 (Dec 15–21): NLU Applications

- **Course:** DataCamp – Natural Language Processing in Python
- **Task:** NER or chatbot using HuggingFace pipeline
- **Output:** Notebook: “*NER / Chatbot Demo*”

✅ Deliverables (Phase 2):

- 4 coding notebooks (ML, CNN, RNN, Transformer/NER)
 - 1 Kaggle submission
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Phase 3 (Dec 22, 2025 – Feb 28, 2026): MLOps + Leadership Prep (9 weeks)

Goal: Deployment, system design, interview prep

◆ Week 17 (Dec 22–28): MLOps Basics

- **Course:** [Coursera – Introduction to ML in Production](#)
- **Task:** Short report on real-world MLOps case
- **Output:** PDF summary

◆ Week 18 (Dec 29 – Jan 4): Deployment

- **Course:** DataCamp – Model Deployment in Python
- **Task:** Deploy ML model with Flask or FastAPI
- **Output:** Deployed model (local / cloud)

◆ Week 19 (Jan 5–11): Monitoring & Maintenance

- **Course:** DataCamp – Monitoring Machine Learning Models
- **Task:** Design ML monitoring pipeline (concept + code demo)
- **Output:** Notebook + Architecture diagram

◆ Week 20 (Jan 12–18): System Design

- **Study:** AI system design interviews (recommendation system, spam filter)
- **Task:** Draw 2 architecture diagrams
- **Output:** PDF: “*AI System Design Notes*”

◆ Week 21 (Jan 19–25): Resume & Storytelling

- **Task:** Rewrite Resume + LinkedIn → AI/ML leadership focus
- **Output:** Updated resume + LinkedIn

◆ Week 22 (Jan 26 – Feb 1): Behavioral & Strategy Qs

- **Practice:** “How would you build an AI strategy?” / “Responsible AI?”
- **Output:** Doc: “*AI Leadership Q&A*”

◆ Week 23 (Feb 2–8): Technical Interview Review

- **Revise:** Bias-variance, cost functions, regularization, CNN/RNN/Transformer
- **Output:** Personal “Interview Cheat Sheet” PDF

◆ Week 24 (Feb 9–15): Mock Interviews

- **Task:** 2–3 mock sessions (peer/mentor/recorded)
- **Output:** Self-review notes

◆ Final 2 Weeks (Feb 16–28): Capstone + Buffer

- **Capstone Project:** End-to-end ML or NLU pipeline → deployed + monitored
- **Output:** Final portfolio project + resume-ready writeup

✓ Deliverables (Phase 3):

- Deployed ML model
- Monitoring pipeline design
- AI System Design PDF
- Updated Resume + LinkedIn
- Capstone project



Final Outputs After 6 Months

- 7+ coding notebooks (Math, ML, DL, NLU, MLOps)
 - 1 Kaggle competition submission
 - 1 Deployed ML project (Flask/FastAPI)
 - Capstone (end-to-end pipeline with monitoring)
 - Updated Resume + LinkedIn
 - Interview prep PDFs (math, design, behavioral)
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👉 This now has **dates + links + tasks + outputs**.

Would you like me to also **generate a clean calendar view (like a Google Calendar/Excel sheet)** so you can track week-by-week progress visually?

You said:

yes please create both Google Calendar and Excel sheet